

Flare Rise Time Sets Recurrence Clock

QPEs and nuclear transients share a tight timescale relation spanning hours to years.

arXiv:2609.08737 — QPEs and recurrent nuclear transients unified by timescale scaling.

arXiv:2609.08737 doi:10.48550/arXiv.2609.08737



Background

Quasi-periodic eruptions and recurrent nuclear transients are repeating, high-amplitude flares from galactic nuclei. Their recurrence times range from hours to years, but the physics setting these clocks is unclear.

Recurrence time scatters widely with black hole mass, so what actually governs when these flares repeat?

What this paper is selling

Compiling QPEs and recurrent nuclear transients, the authors find recurrence time scales almost linearly with flare rise time. This tight relation links systems separated by orders of magnitude in mass and timescale.

Key results to remember

- Recurrence time scales with black hole mass, but with huge 0.95 dex scatter.
- Recurrence time tracks rise time nearly linearly, scatter only 0.29 dex.
- One relation spans hours-long QPEs to years-long nuclear transients.

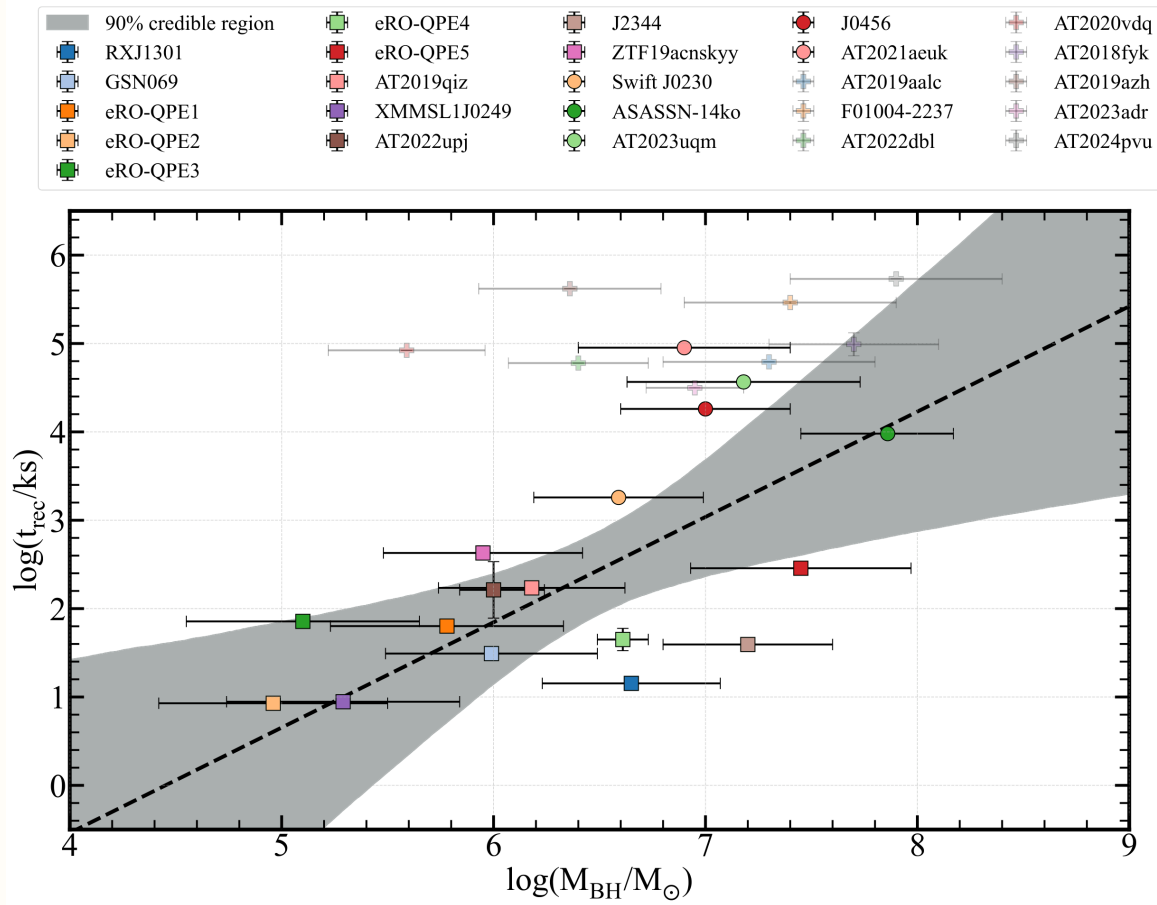


Figure 1 | Black hole mass versus recurrence time: a positive trend, but with scatter too large to be predictive.

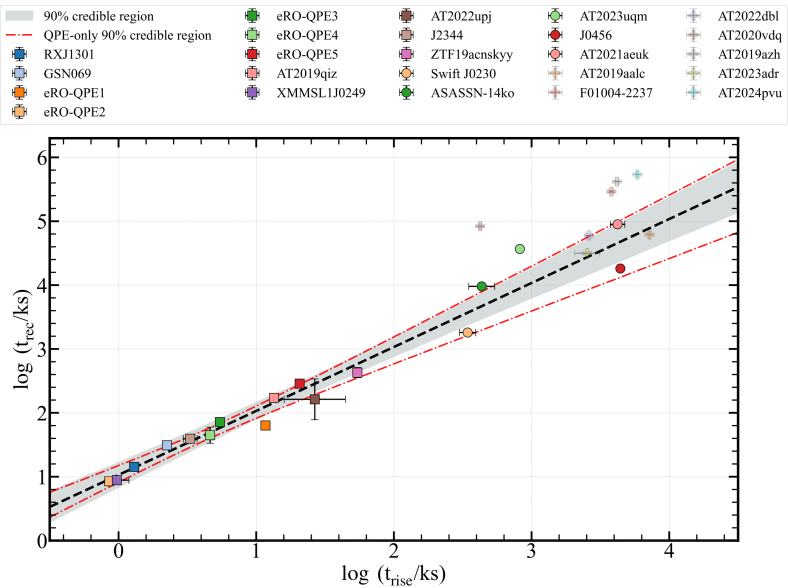


Figure 2 | Rise time versus recurrence time: a strikingly tight, nearly linear relation across all systems.